ParaliSetu — Business Model Canvas

AI-Driven Stubble Aggregation, Baler Fleet Optimization & Escrow Marketplace

Hackathon: Hack for Social | **Track:** Environment & Sustainability | **Geography:** Punjab & Haryana Hub | **Code:** 100% Green Tests

8 KEY PARTNERS

INDUSTRIAL BIO-ENERGY OFFTAKERS

- **Bio-CNG Plants (SATAT):** Verbio Sangrur, EverEnviro, Reliance Bio-Energy.
- **2G Ethanol Refineries:** IOCL Panipat 2G Bio-Refinery.
- Biomass pelletizers & industrial boilers co-firing 5-7% agro-pellets.

EQUIPMENT & TRANSPORT POOLS

- **Custom Hiring Centres (CHCs):** Subsidized pools of tractors, square balers & rakes.
- Local tractor-trolley fleet owner networks.

6 KEY ACTIVITIES

- **PostGIS Clustering:** Real-time grouping of 10-15 fragmented smallholders into 35-acre runs.
- **Vernacular Voice AI:** Dialect-accurate Punjabi & Hindi speech-to-booking pipeline

7 KEY RESOURCES

- **Proprietary Software IP:** PostGIS spatial matching engine, offline voice parser & Next.js portal.
- **PAU Agronomic Formula:** Calibrated yield factors (2.0 tonnes/acre baler recovery)

2 VALUE PROPOSITIONS

FOR SMALLHOLDER FARMERS (1-4 ACRES)

- **Free Field Clearance:** Stubble cleared in 48-72h before wheat sowing window.
- **Guaranteed Cash Earnings:** ₹1,200-₹1,800/acre net income (₹650/tonne direct payout).
- **Zero Pollution Penalties:** Avoid ₹5,000-₹15,000 CAQM fines + verified No-Burn Certificate.

FOR BALER & FLEET OPERATORS

- **Contiguous 30-50 Acre Runs:** Eliminates road transit diesel wastage ("deadhead miles")

4 CUSTOMER RELATIONSHIPS

- **Voice-First Simplicity:** Natural speaking in Punjabi/Hindi; zero tech literacy barrier.
- **Dharamkanta Scale Trust:** Weight verified at local village scale before truck dispatch

3 CHANNELS

- **Flutter Mobile App:** 17MB lightweight Android app with offline speech-to-text fallback.
- **Next.js Operations Web Portal:** Scale ticket logging, buyer demand setting & audit view

1 CUSTOMER SEGMENTS

PRIMARY BENEFICIARIES

- **Small & Marginal Farmers (1-4 Acres):** Trapped in the 15-day harvest deadline. Cannot afford machinery purchase. Pays ₹0

PAYING COMMERCIAL CUSTOMERS

- **Bio-CNG (CBG) Plants:** Under MoPNG's **SATAT mandatory blending obligation (1% to 5%)**.
- **2G Ethanol Refineries & Power Plants:** Legally mandated to co-fire 5-7% biomass.

COMMERCIAL SUPPLY OPERATORS

- Custom Hiring Centre (CHC) baler

9 COST STRUCTURE

TECHNOLOGY & PLATFORM OVERHEAD (FIXED)

- **Cloud Infrastructure:** Supabase PostgreSQL/PostGIS spatial DB, Next.js hosting: ₹35,000/month.
- **Voice AI & SMS:** Whisper/Gemini ASR tokens + transactional SMS OTPs: ₹0.27 per voice intake.
- **Satellite Pipeline:** Copernicus Sentinel-2 SWIR NBR automated processing overhead.

FIELD OPERATIONS & QUALITY CONTROL (VARIABLE PEAK)

- **Ground Coordinators (Kisan Mitras):** 10 village cluster leads during 35-day harvest peak (₹1.50L).
- **Moisture Quality Kits:** Handheld digital pin probe moisture meters (₹2,500/unit).
- **Banking Escrow & Compliance:** Third-party weighment auditing & automated banking API fees.

5 REVENUE STREAMS & COMMERCIAL VIABILITY

- **B2B Offtake Take-Rate (Primary):** ₹150 / Tonne (5.8% facilitation fee on ₹2,600 delivered gate price, paid 100% by industrial buyers; **0% fee for farmers**).
- **Baler Optimization Route SaaS:** ₹40 / Tonne software fee for pre-clustered contiguous runs.
- **Enterprise ESG & Scope 3 Data:** Verified supply chain traceability subscription for Oil Marketing Companies (OMCs).
- **Financial Breakeven:** Cash break-even achieved at **20,000 Tonnes (~10,000 acres in 1**

ParaliSetu Framework: Grounded in PAU Ludhiana agronomic recovery data & MoPNG SATAT Blending Mandate.

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